

WiGiTS Oncoanalyser v3.0 Release Notes

Overview

The v3.0 pipeline release has these key improvements:

- Ultima & Roche Axelos sequencing support
- Various extensions to driver reportability
- Call germline amplifications
- Adapt Orange report to driver reportability changes
- Improvements to noise detection and filtering and purity fitting in tumor only mode

Resource File Updates

Tool / Stage	Description
Fusion KB	Added ~75 new known and promiscuous fusions including TCR enhancer rearrangements Renamed 'IG' types to 'ENHANCER'
Driver gene panel	New columns: reportLoh, reportGermlineAmplification, reportHighExpression, reportLowExpression, reportNovelSpliceJunction Updated: UGT1A8 renamed to UGT1A1 CCND2 & CCND3: ReportDisruption=TRUE CTNNA1: enable germline reporting EGFR: enable germline reporting KRAS: enable germline reporting MET: reportNonsense=TRUE GREM1: constrict settings to ONCO relevant and set to VARIANT_NOT_LOST/WILDTYPE_LOST AIP: set to VARIANT_NOT_LOST/WILDTYPE_LOST MBD4: set to VARIANT_NOT_LOST/WILDTYPE_LOST RHBDF2: constrict settings to ONCO relevant and set to VARIANT_NOT_LOST/WILDTYPE_LOST Removed: IGLL5, HNRNPA2B1, JMJD4, RARG, TRIM51, LMBRD2, SEMG2, TPTE, ITGAV, C21orf91, LRRN3, USP22, WWC3, KRTAP5-5, KANSL1, KRT5, NIPBL, OR11H1, OR4N2, PDYN, RACGAP1, SMTNL2, SPTAN1, ZFX Added: MAPK1, XRCC2, GREM1, AIP, MBD4, RHBDF2, CSF1, HMGA2
PON files	Added Ultima and SBX small variant PONs
MSI model	Added WGS coefficients for all seq-technologies
MSI site list	New down-sampled microsatellites file which has more data in sparse rows and as used in Redux for microsatellite error calculations (and per-panel MSI predictions in targeted mode)
Small variants	MSH2 variant to germline blacklist VCFs
Amber sites	Added Gnomad frequency
Germline AMPs and DELs	Regenerated and added germline amplifications from WGS cohort

Tool Updates

See tool releases and READ-ME for full details.

Tool	Version	Release Notes and Key Changes
Amber	4.3	Functional: - filter out sites with $\geq 15\%$ reads with low base or map qual - base qual threshold 30 -> 13 - include Ultima low-qual bases in filter count - targeted panel contamination model: LOD depth dependent but 0.5% to 1.0% for a typical deep panel). Min BAF exported by Amber raised from 2.5% to 4.0% but will be lowered conditionally if evidence of a low BAF copy number-related peak is identified (see below point) Bugs: - apply tumor filter to sites used in contamination Technical: - use new segmentation routine - no contamination VCF file output
Bam tools	1.6	Functional: - capture off target read counts, restrict coverage to target regions by default without 'only_target' config Technical: - metrics write gene and exon coverage with upper-case fields - rename exon_medians.tsv to exon_coverage.tsv - add mean depth and proportion of bases above depths metrics to exon coverage file
Cider	1.2	Functional: - annotate V/D/J genes by matching to the original IMGT - output somatic hypermutation status. Adapted previous gene percentage identity output (https://pmc.ncbi.nlm.nih.gov/articles/PMC7248390/) - new filter V_J_OVERLAP for annotated gene which align in the wrong order Technical: - use read mates in assembly
Cobalt	2.0	Functional: - normalisation: invalidate regions with significantly less (0.15) enrichment than neighbouring - normalisation: dampen percentile towards median - normalisation: optional use of WGS copy-number percentiles file instead of median Technical: - use new segmentation routine
Esvee	2.0	Functional - SBX and Ultima support - consensus building and read assessment uses new mismatch penalty scheme - expand extension mismatch penalty lengths: 1-5 = 0, 6-15 = 0.2, 16-49 = 1, 50-99 = 2, 100+ = 3 - panel: soft-filter off-target variants - ensure PMS2 with low map qual passes alignment filters, including selection of alt alignment - assembly can be marked as LINE if it has half or more valid reads marked as line - allow mismatches in line test ref bases - mark line reads only if ref base A/T repeats ≤ 20

		- read ref base check matching seed multiple times requires non-low-base matches - stop ref base sequencing building if majority of reads are soft-clipped - trim adapter soft-clip bases from 3' end - handle multiple supp data to aid remote region linking - Caller: 3' strand-bias filter applies to both breakends - Prep: filter supps with low map qual (<10) and trimmed align length < 40 - require less than 1/3 bases as low-qual errors to add junction read, capture low-qual mismatches prior to consensus base determined Technical - filter phase groups if assembly count exceeds 1k: - take the top X assemblies only, scored by longest extensions, lowest mismatch ratio and highest frag counts - Prep: depth tracking drops window read counts and window size 1000 -> 200 - Prep: reworked junction finding routines Bugs - fixed assembly facing link bug - will occasionally lead to more facing links being found - phase set building after unmapped extension factor in existing primary link assemblies - Prep: always capture both indel junctions
Isofox	2.0	Functional: - report on gene expression based on cohort comparisons and driver gene panel - trim 3' adapter sequence from paired reads - panels: restricted gene IDs only used to limit expression output based on TPM normalisation, without limiting fusion detection Technical: - writes TSV files instead of CSV - added chimeric excluded regions - write passing alt-SJs if match known exon del-dup - alt splice junction write transcript, exons and a passing file using driver gene panel reportable status - fusion write common known fusion type, transcript and exon from canonical if found otherwise any other
Lilac	2.0	Functional: - support SBX and Ultima - support for 4 MHCII genes: DPA1, DPB1, DQA1, DQB1 - specified indel PON N bases, allowing correct identification of deletes - doubles unmatched indel, AA and haplotype warning levels for panels Output: - write Genes column for key output files to use in Orange and Compar
Linx	2.3	Functional: - deletion bridge check for adjacent now considers orientation - mark non-disruptive if both breakends in same transcript are 3'UTR, or one for DUP, somatic dup-dels in 3'UTR can be disruptive - required 1mb distance for enhancer prom fusions, check known exons if set to be high impact prom - highly impact exon fusions ignore chain termination and downstream distance rules - enhancer fusions without 3' strand can be upstream or coding, within 50K of gene - known fusions DB: 'IG' changed to 'ENHANCER', allow orientation to be 0 meaning not checked

		Technical: - reportable vis plots use simpler naming convention - removed fusion fields only for patient reporter since can be derived from breakends Output files: - breakend: added vcflid and coords (chr:pos:orientation) - breakend: reportedDisruption -> reportedStatus - breakend: removed type, chromosome, orientation, strand, chrBand, junctionCopyNumber - sv-annotation: added type, vcflidStart, vcflidEnd, coordsStart, coordsEnd - fusion: added fivePrimeVcflid, threePrimeVcflid, fivePrimeCoords, threePrimeCoords - fusion: removed geneStart, geneContextStart, geneTranscriptStart, geneEnd, geneContextEnd, geneTranscriptEnd, junctionCopyNumber - cohort files written as TSVs
Orange	5.0	Functional: - various changes to germline and somatic tables - reworked front page - removed potentially interesting events - hide immune section - show cancer type as 'NOT SPECIFIED' if missing from config - RNA events only shown if reportable - show driver candidates in grey - only include events linked to driver events for all JSON output - removed germline to somatic logic
Pave	1.9	Functional: - use PON multi status for SBX and Ultima filter
Purple	4.4	Functional: - restrict fitting hotspots to 5% VAF - added LIKELY_DIPLOID GermlineStatus for diploid percentage 0.7-0.85 and 1.15-1.3 - TMB and TML raise min VAF of 0.02 -> 0.05 - clustering only uses Amber BAF PCFs if follows another one and not at start or end of an arm - check depth window count and SV-bounds for germline amp-dels - use BAF fit if highly diploid and high purity if BAF within specified limits - show driver genes on Circos plot - aneuploidy detection to impact somatic fit - assign the new segment support type "EXCL" to new fragments when splitting segments by excluded regions - call germline AMPs as well as DELs - segment around excluded IG regions - find driver for LOH events with zero likelihood - find drivers splice region without gene impact and zero likelihood Targeted panels: - use MS model prediction to set MSS/MSI - adjusted targeted TML calc - use chrX region size to determine Amber gender - removed deleted genes logic Bugs: - fix plot fail when unable to determine maxPloidy. Replace deprecated ggplot 'size' arg with 'linewidth'

		Output: - dropped chimerism field from QC - germline amp-del specifies if partially covers gene
QSee	1.0	New component added to produce comprehensive quality control report
Redux	2.0	Functional: - support for SBX and Ultima - jitter UMI duplicate collapsing allows no edit distance Technical: - runs BQR instead of Sage - refactored jitter duplicate and poly-G UMI collapsing - optimised read cache for intense regions (> 100K reads per 400 bases)
Sage	5.0	Functional: - support for SBX and Ultima - use Redux MSI prediction to set MS status for jitter params - read edge distance and event penalty calculations now scale with read length - Illumina, candidate-creating reads must have no low qual bases across variant base(s) (SNV/MNV) or variant core (indel) - edge distance penalty in MQF test not applied to long indels isolated from other multi-base indels - reads with uncertain or medium qual bases are not used in jitter calculations - minStrongSupport filter now only considers reads with high base quals for SNV/MNVs - maxGermlineVAF filter conditionally relaxed for MSI indels in PANEL tier - 1.5x error rate jitter threshold for hotspots only applied if jitter ratio > 3% (prev 2.5%) - dedupIndel logic works on germline-filtered indels - all genes in driver gene panel get 'tier=PANEL' regardless of reportable flag, but excluding UTR regions - min repeat count for 'double jitter' in MSI samples increased from 11 to 21 - germline variants in PANEL tier are no longer hard filtered Technical: - removed BQR generation, now loaded from Redux - continue with invalid read contexts on non-human chromosomes and/or low positions Output: - VCF annotations renamed: ABQ -> ARCBQ, SB -> FSB, RABQ -> ASBQ, AMQ -> ARMQ, AMMQ -> AFMQ, AMBQ -> AFBQ - VCF added: RSB, TINC_RECOVERED
Teal	1.4	Technical: - support for unpaired reads
Wisp	1.3	Functional: - Ultima has specific qual and edge distance filters

Configuration Changes

Tool	Release Notes and Key Changes
Amber	- specify sequencing_type
Esvee	- specify sequencing_type
Isofox	- add alt_sj_cohort_file - add gene_distribution_file - add driver gene panel

Lilac	- specify sequencing_type - targeted_panel
Orange	- specify sequencing_type - added qsee dir - added output_id - add primary_tumor_location - specify panel_name for display - remove signatures etiology, driver gene panel, Ensembl data cache, cohort_mapping_tsv, cohort_percentiles_tsv - removed RNA: alt_sj_cohort_file, gene_distribution_file, cohort_percentiles_tsv - removed limit_json_output, convert_germline_to_somatic
Pave	- specify sequencing_type
Purple	- amber -> amber_dir - cobalt -> cobalt_dir - add redux_tumor_dir - to load MS production file for panels - germline_del_freq_file -> germline_amp_del_freq_file - removed targeted panel ratio file, replaced by optional config: target_tmb_ratio, target_tml_ratio, target_coding_factor
Redux	- specify sequencing_type - MSI mode error rates and coefficients: 'msi_model_error_rates' and 'msi_model_coefficients' - support UMI-type KAPA - support UMI-type MSK UMIs (variable length)
Sage	- specify sequencing_type
Teal	- purple -> purple_dir - cobalt -> cobalt_dir
Wisp	- specify sequencing_type

Technical Changes

Tool	Performance Characteristics
Amber	Removed R CN dependency
Cobalt	Removed R CN dependency
All tools	Support for SBX & Ultima